Institute of Geographical Information Systems
School of Civil & Environmental Engineering

National University of Sciences and Technology, Islamabad, Pakistan

PBL Activity	
Semester: 3rd BE-2025	**GIE-203 Digital Mapping and Image Processing**
Time Allowed: 1 week	**Marks/ Score: 10**

Topic: Use of Vegetation Indices for Mapping Green Vegetative Cover (CLO2, PLO4)

Objectives: To enable students to look for freely available satellite imagery, acquire relevant season images, and apply different vegetation indices for mapping wheat-cropped areas.

Relevance to Real Life (Practical Application): This problem-based learning (PBL) activity will enable students to learn about the geospatial data portals, which provide freely available satellite imagery from several remote sensing satellite systems. It will help them select appropriate imagery for mapping rice-cropped areas at different temporal resolutions. The mapped information through remote sensing can help to estimate the rice-cropped areas and may be used for early yield crop estimation.

Problem Statement: Your client has assigned you a study area, which is majorly an agriculture area. The client needs the rice-cropped areas mapped within the study area using freely available remotely sensed data. You must download freely available remotely sensed imagery of the study area of the appropriate season from Landsat 8-9, Sentinel, or whichever satellite system you consider appropriate and apply any indices based technique better for mapping rice-copped areas.

Task: As an image analyst, you must map/extract and provide the client with the best and most accurate information. The client may be interested and looking for the following response.

Required Response:

a. **Apply** appropriate vegetation index for mapping total green vegetative cover within the study area
b. **Classify** Rice-cropped areas from total vegetative cover
c. **Compute** of the cropped areas (Acre/ Hectares)
d. **Relate** the health, distribution and yield of the wheat cropped areas

Institute of Geographical Information Systems
School of Civil & Environmental Engineering

National University of Sciences and Technology, Islamabad, Pakistan

Deliverables: Share the output in the form of maps and a written report covering important aspects of the operations you performed, and the analysis of the outputs. The report must cover the following aspects:

a. Brief of the assigned task and the information about selected/ used data
b. Identification of the appropriate processing technique
c. Purpose of the applied operations
d. Evaluation and analysis of the outcomes

PBL Report as per Format below.

Due Date: Tuesday Nov 18, 2025

Institute of Geographical Information Systems
School of Civil & Environmental Engineering

National University of Sciences and Technology, Islamabad, Pakistan

Format Report

Introduction

Study Area

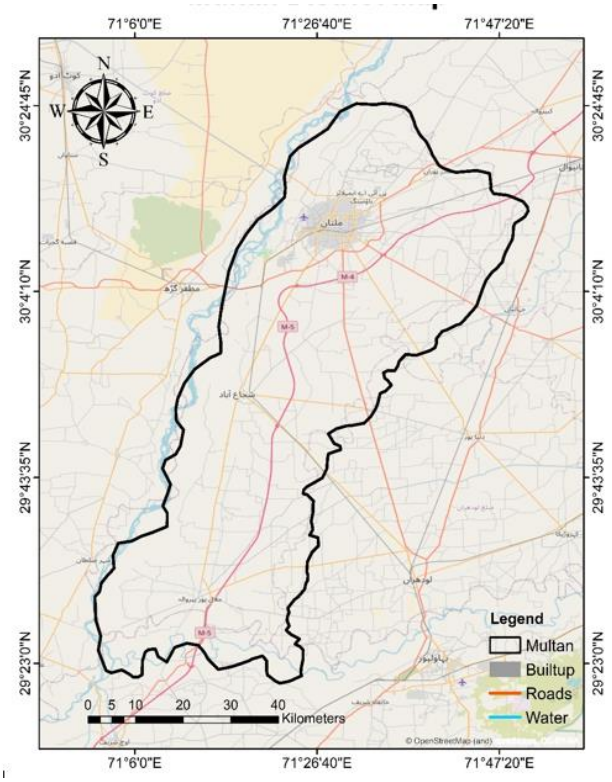


Figure.1 Study Area Map- District XXXX

Institute of Geographical Information Systems
School of Civil & Environmental Engineering

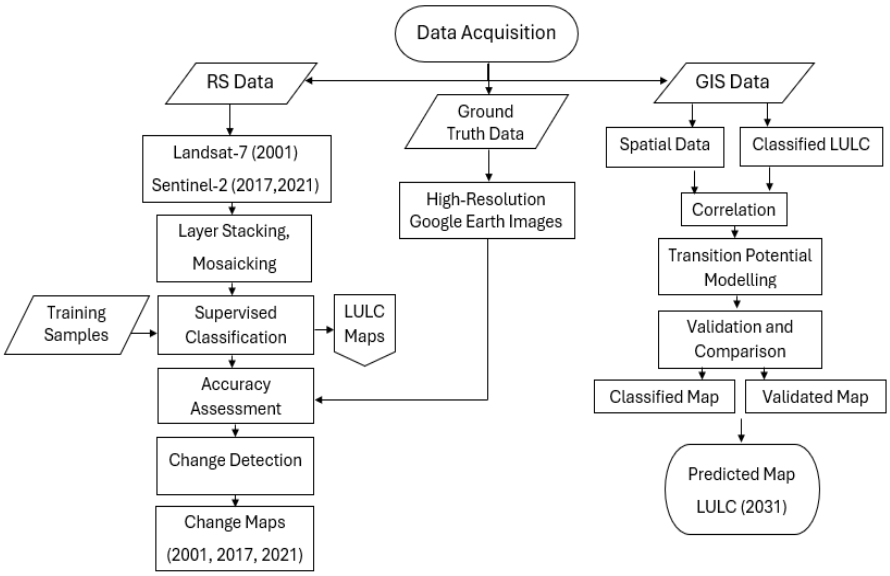
National University of Sciences and Technology, Islamabad, Pakistan

Dataset

Table 1: Dataset used in the study along with specifications and sources

Data	Specification	Sources
Satellite Data	Landsat-7 ETM (2001) 30m*30m spatial resolution	USGS Earth Explorer
Satellite Data	Sentinel-2 L1C (2017, 2021) 10m*10m spatial resolution	Copernicus Open Access Hub
Agriculture Data	Crop Yield, Crop Area Coordinates	Cotton Crop Research Institute Multan

Methodology



Results

Table.1 XXXXXX

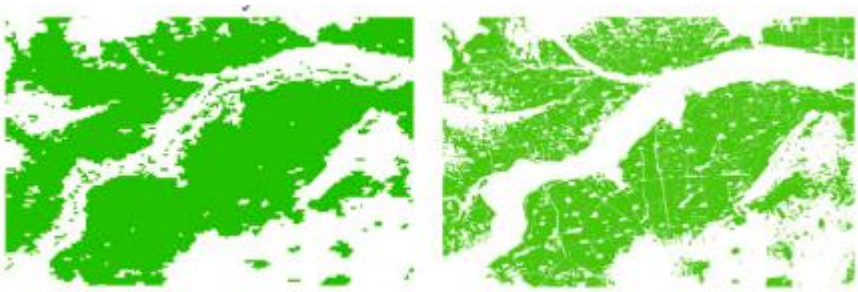


Figure.1 XXXXX

Conclusions

References